8. Expectation, Variance, Covariance, Independence

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```r
#Purpose: What do these concepts look like in code?
#Clear environment
rm(list = ls())
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.2     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.4
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(roxygen2)
library(plotly)
```

```
##
## Attaching package: 'plotly'
##
## The following object is masked from 'package:ggplot2':
##
##     last_plot
##
## The following object is masked from 'package:stats':
##
##     filter
##
## The following object is masked from 'package:graphics':
##
##     layout
```

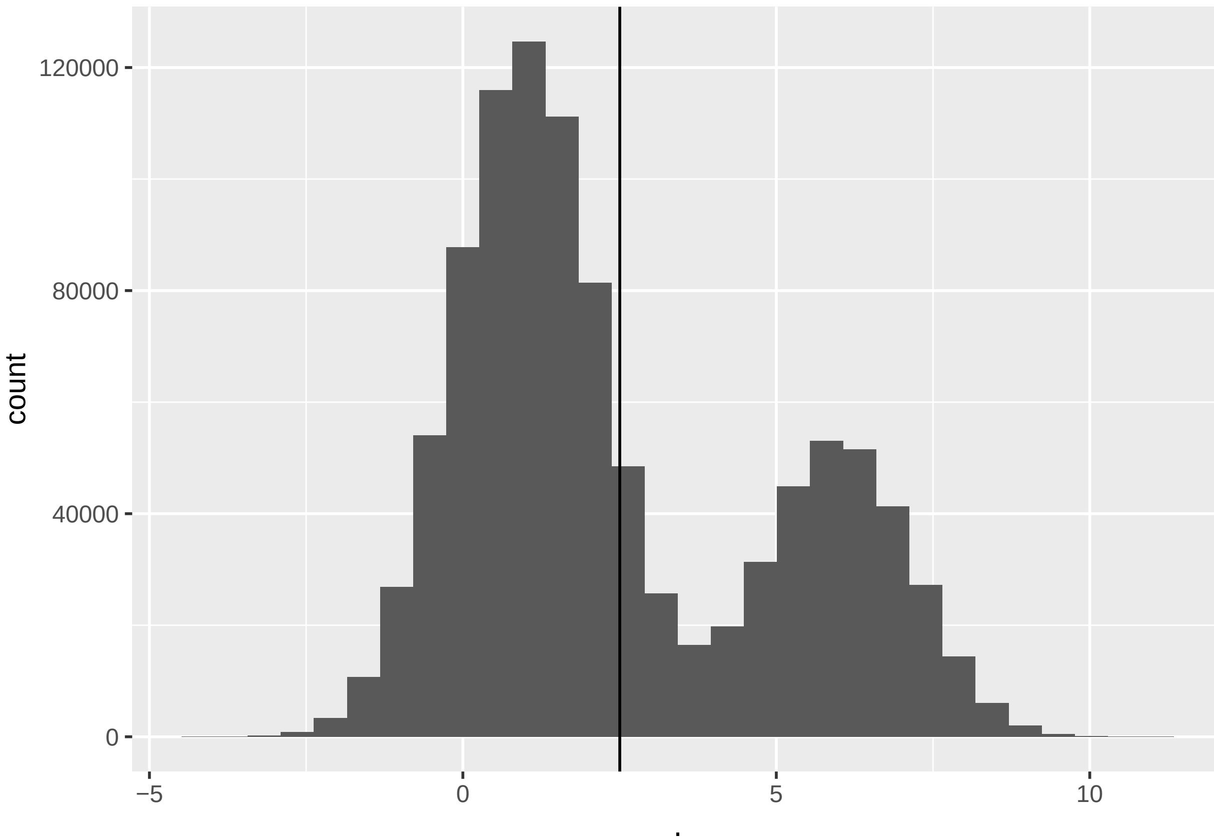
```r
library(rgl)

set.seed(60637)

#----Expectation----
#Recap: what is expectation?
```

```r
#We cannot observe it directly with data, but we can estimate!

#So far, we've talked about distributions in a theoretical sense, looking at different properties of ra
#We don't observe random variables; we observe realizations of the random variable. These realizations
#We'll spend more time in the intro class talking about this from the standpoint of *estimands*, *estim

#If we draw from a distribution, even an unknown one, is the sum of the values times their probability.

#When we have data, our best guess simplifies to a mean, because the probability a value is in the data
#This makes more sense when we consider any data as sampling from some (super)population, assuming no b

#What's the sample expectation of this weird distribution?
distribution<-rbinom(1000000,1,.3)*5 + runif(1000000,0,1)*2 + rnorm(1000000,0,1)%>%
  as.data.frame()


#simple as:
mean(distribution$.)
```

```
## [1] 2.502987
```

```r
#balance the shape!
ggplot(data = distribution,
       aes(x = .))+
  geom_histogram() +
  geom_vline(xintercept = mean(distribution$.))
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```




```
#The sample expectation, in the discrete data world (as opposed to the continuous, distribution ideal)
#when you think about it that way, some nice properties come out. We'll prove each if we have time, but

#Exercise 1: Demonstrate that
```

1. Expectation of a constant is a constant
$$E(c) = c$$
2. Constants come out
$$E(cg(Y)) = cE(g(Y))$$
COME BACK TO THIS:
3. Expectation is Linear
$$E(g(Y_1) + \cdots + g(Y_n)) = E(g(Y_1)) + \cdots + E(g(Y_n)),$$
regardless of independence
4. Expected Value of Expected Values:
$$E(E(Y)) = E(Y)$$
(because the expected value of a random variable is a constant)

```
#----sample variance----
#We know where a distribution (or sample) is centered. But what about its spread?
#In other words, how far from the average is the average observation?

#There are different ways to conceptualize this, but we (and everyone else) will use the following:
```

$$\mathrm{Var}(Y) = E[(Y - E(Y))^2] = E(Y^2) - [E(Y)]^2$$

```
#Exercise 2: prove it!

#in R, it's easy
var(distribution$.)
```

```
## [1] 6.577084
```

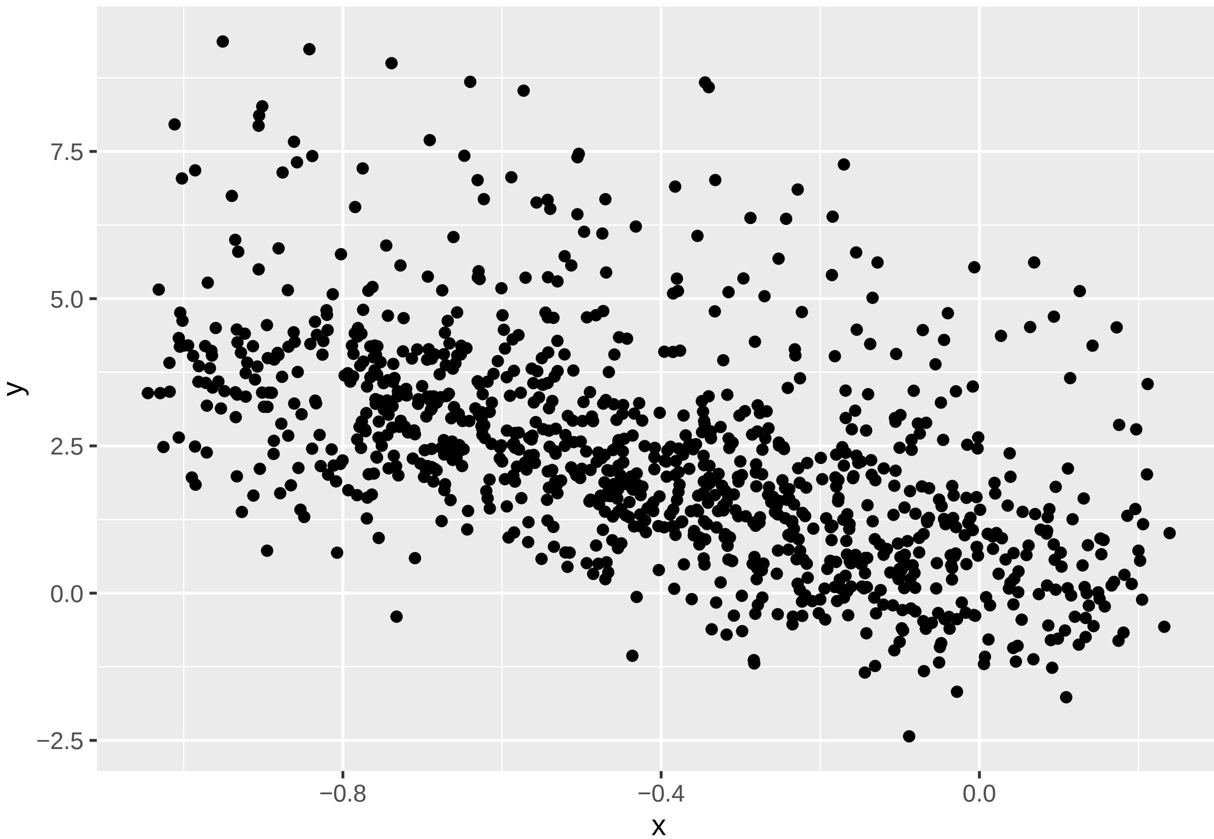
```
#Because sample variance has a squared term, its properties are not as straightforward as sample expect


#Exercise 3: Calculate the sample variance of
```

$$f(x) = \begin{cases} \frac{3!}{x!(3-x)!}(\frac{1}{2})^3 & x = 0,1,2,3 \\ 0 & otherwise \end{cases}$$

```
#Hint: First calculate $E(Y)$ and $E(Y^2)$



#----Covariance----

#The covariance measures the degree to which two random variables vary together;
#if the covariance between $X$ and $Y$ is positive, X tends to be larger than its mean when Y is larger
```

$$\mathrm{Cov}(X, Y) = E[(X - E(X))(Y - E(Y))]$$

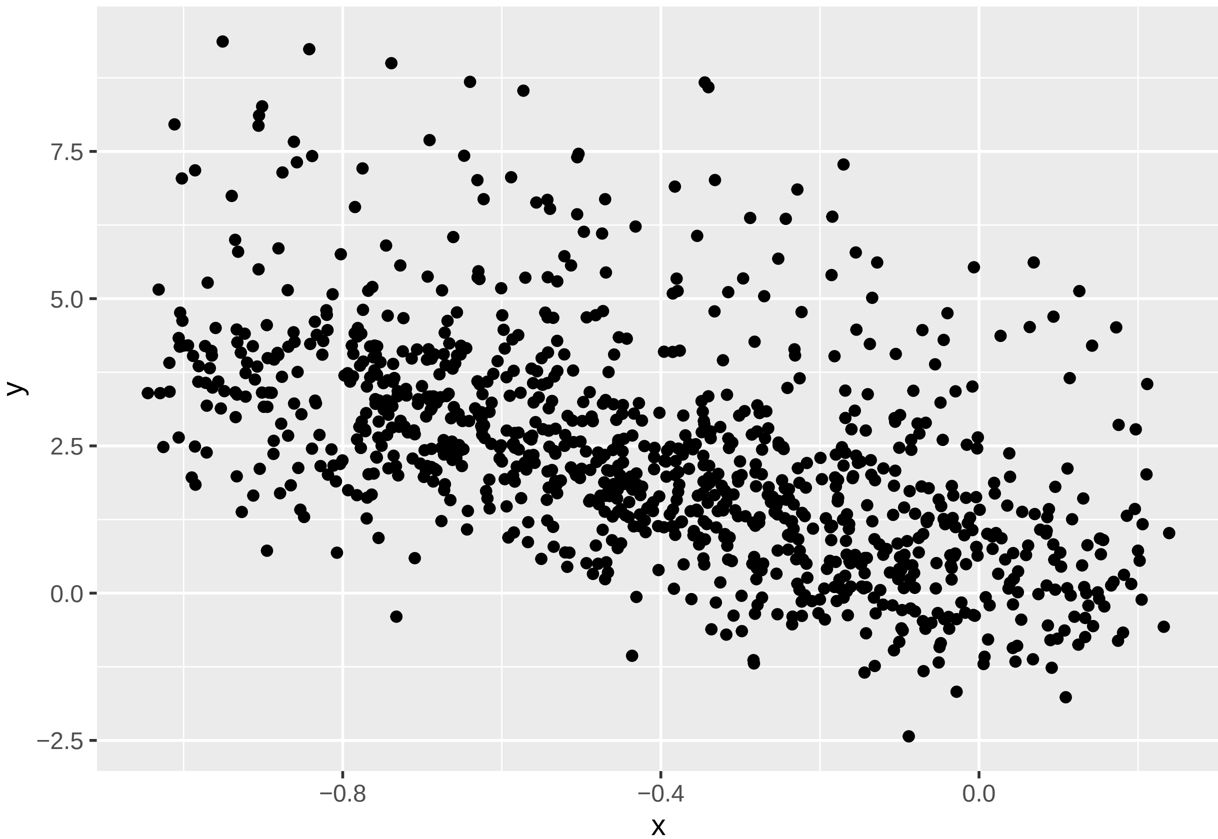
```
#What's the covariance of the following sample,from a weird joint distribution:
n<-1000
y<-seq(from = 1/250, to = 4, by =1/250) +rnorm(n = n) + rbinom(n= n, size =1,.1)*4
x<-rweibull(n, shape = 2, scale = .1) - rep(c(.1,-.1),500) - seq(1/n,1, by = 1/n)

ggplot()+
  geom_point(aes(x = x,
                 y = y))
```



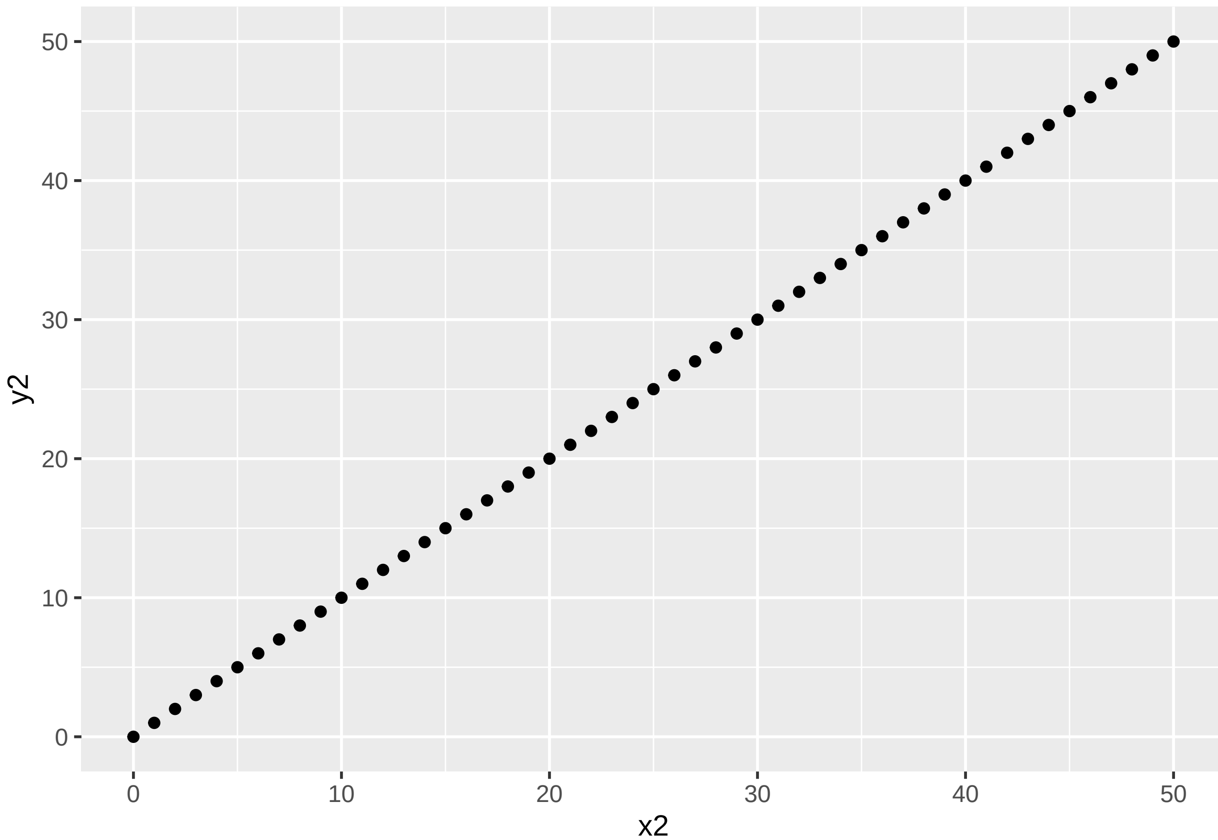

```
#Hard to do analytically from distributions. Easy to sample and calculate an estimate.
cov(x,y)
```

```
## [1] -0.320684
```

```
#RETURN TO THIS:
#Notice that covariance scales unhelpfully...
x1<-seq(0,100)
y1<- seq(0,100)

ggplot()+
  geom_point(aes(x = x,
                 y = y))
```




```r
#X and Y have a similar relationship here,  no?
x2<-seq(0,50)
y2<-seq(0,50)

ggplot()+
  geom_point(aes(x = x2,
                 y = y2))
```




```r
cov(x1,y1)
```

```
## [1] 858.5
```

```r
cov(x2,y2)
```

```
## [1] 221
```

```r
#To avoid this problem we use correlations, which scale between -1 and 1:
```

$$\mathrm{Corr}(X,Y) = \frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X)\mathrm{Var}(Y)}} = \frac{\mathrm{Cov}(X,Y)}{SD(X)SD(Y)}$$

```r
cor(x1,y1)
```

```
## [1] 1
```

```r
cor(x2,y2)
```

```
## [1] 1
```

```
#Properties of variance and covariance, for your reference:
```

1. $\text{Var}(c) = 0$
2. $\text{Var}(cY) = c^2\text{Var}(Y)$
3. $\text{Cov}(Y, Y) = \text{Var}(Y)$
4. $\text{Cov}(X, Y) = \text{Cov}(Y, X)$
5. $\text{Cov}(aX, bY) = ab\text{Cov}(X, Y)$
6. $\text{Cov}(X + a, Y) = \text{Cov}(X, Y)$
7. $\text{Cov}(X + Z, Y + W) = \text{Cov}(X, Y) + \text{Cov}(X, W) + \text{Cov}(Z, Y) + \text{Cov}(Z, W)$
8. $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$

```
#Exercise 4: Groups prove one each!

#Exercise 5: Find the expectation and variance.
#Now find the sample expectation and sample variance
```

Given the following PDF:

$$f(x) = \begin{cases} \frac{3}{10}(3x - x^2) & 0 \leq x \leq 2 \\ 0 \quad otherwise \end{cases}$$

```
#----Conditionals----
#If time, show
# Load Titanic data
data(Titanic)
titanic_df <- as.data.frame(Titanic)

# Expand the data (each row represents Freq number of people)
expanded_titanic <- titanic_df[rep(row.names(titanic_df), titanic_df$Freq), 1:4]


#What's the overall chance of survival? P(Survived)
expanded_titanic$survived_bin<-ifelse(expanded_titanic$Survived == 'Yes',1,0)
survived<-sum(expanded_titanic$survived_bin)
survived/nrow(expanded_titanic)
```

```
## [1] 0.323035
```

```
mean(expanded_titanic$Survived =='Yes')
```

```
## [1] 0.323035
```

```
#P(Survived| Class = 2nd)
expanded_titanic%>%
  filter(Class == '2nd') %>%
  summarise(prob_survived = mean(Survived == 'Yes'))
```

```
##   prob_survived
## 1     0.4140351
```

```r
#Exercise 6: Which subgroup had the lowest chance of survival?
  #consider using prop.table
?prop.table()
```

```
## starting httpd help server ... done
```

```r
attach(expanded_titanic)
table(Survived,Class,Sex,Age)%>%
  prop.table()
```

```
## , , Sex = Male, Age = Child
##
##         Class
## Survived          1st          2nd          3rd         Crew
##      No  0.0000000000 0.0000000000 0.0159018628 0.0000000000
##      Yes 0.0022716947 0.0049977283 0.0059064062 0.0000000000
##
## , , Sex = Female, Age = Child
##
##         Class
## Survived          1st          2nd          3rd         Crew
##      No  0.0000000000 0.0000000000 0.0077237619 0.0000000000
##      Yes 0.0004543389 0.0059064062 0.0063607451 0.0000000000
##
## , , Sex = Male, Age = Adult
##
##         Class
## Survived          1st          2nd          3rd         Crew
##      No  0.0536119945 0.0699681963 0.1758291686 0.3044070877
##      Yes 0.0258973194 0.0063607451 0.0340754203 0.0872330759
##
## , , Sex = Female, Age = Adult
##
##         Class
## Survived          1st          2nd          3rd         Crew
##      No  0.0018173557 0.0059064062 0.0404361654 0.0013630168
##      Yes 0.0636074512 0.0363471149 0.0345297592 0.0090867787
```

```r
survival_prop<-expanded_titanic%>%
  group_by(Age,Class,Sex)%>%
  summarise(mean(survived_bin))
```

```
## `summarise()` has grouped output by 'Age', 'Class'. You can override using the
## `.groups` argument.
```

```r
expanded_titanic%>%
  group_by(Class)%>%
  summarise(mean(survived_bin))
```

```
## # A tibble: 4 x 2
##   Class `mean(survived_bin)`
```

```
##   <fct>                <dbl>
## 1 1st                  0.625
## 2 2nd                  0.414
## 3 3rd                  0.252
## 4 Crew                 0.240
```

```
#What's the probability of being in 2nd class given that you've survived P(Class = 2nd|Survived)?


expanded_titanic%>%
  filter(Survived == 'Yes') %>%
  summarise(prob_2nd = mean(Class == '2nd'))
```

```
##    prob_2nd
## 1 0.1659634
```

```
  #Johnson et al, 2019...

set.seed(3)
customers <- data.frame(
  age_group = sample(c("18-25", "26-40", "41-55", "56+"), 2000, replace = TRUE,
                     prob = c(0.3, 0.35, 0.25, 0.1)),
  purchase_made = sample(c("Yes", "No"), 2000, replace = TRUE, prob = c(0.4, 0.6)),
  device = sample(c("Mobile", "Desktop", "Tablet"), 2000, replace = TRUE,
                  prob = c(0.6, 0.3, 0.1))
)

#What's the probability of making a purchase?
mean(customers$purchase_made == 'Yes')
```

```
## [1] 0.402
```

```
#What's the probability of making a purchase for those in the youngest category?
mean(customers$purchase_made[customers$age_group== '18-25'] == 'Yes')
```

```
## [1] 0.4212329
```

```
#Exercise 7: Do this with tidyverse

#P(purchase b * P(purchase|age group = 18-25). What does that mean about age and purchase propensity?
  #note, they're not exactly the same. Is that an actual difference or due to chance?



#Continuous Conditional Expectation (Thanks to Andy Eggers!)
#With probability and outcomes, we can take expectations.
  #Note, we've actually been doing expectations in the above, but since the outcomes were binary, proba



dat <- expand_grid(x = seq(-10, 10, length = 300),
                   y = seq(-10, 10, length = 300))
```

```
Sigma <- cbind(c(1, .7), c(.7, 1))
dat |>
  mutate(z = mvtnorm::dmvnorm(as.matrix(dat), sigma = Sigma)) -> datt2

# open3d()
#
# plot3d(datt2, col = "grey50", alpha = .5,
#        xlab = "x", ylab = "y", zlab = "f(x, y)",
#        xlim = c(-3, 3), ylim = c(-3, 3),
#        aspect = c(1, 1, .5))
#
# #what's the expectation? #E[f(x,y)]
# #What's #E[f(x,y)| Y = 1]
# planes3d(a = 1, b = 0, c = 0, d = -1,
#          col = "red",
#          alpha = .5)

n <- 100
y <- seq(-3, 3, length = n)
x <- rep(1, length(y))
z <- mvtnorm::dmvnorm(cbind(x, y), sigma = Sigma)
# lines3d(x = x, y = y, z = z, col = "black", lwd = 2)

#we can do conditional variance, covariance, etc....






#But we won't today.
```